

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.* (BL3, BL4)

Activity Tool : Mock Interview (Low)
Assessment Tool Weightage: 30%

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QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	5
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	5
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	5
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	5
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	5
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	5
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	5
8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	5

9	Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'	BL4 – Analyze	5
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	5

Code of Conduct:

I G Vyshnavi (192525260) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: vyshnavi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively

Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Q1. Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?

Query processing is the process of converting a SQL query into an efficient execution plan and obtaining the required result.

The main steps are:

1. **Parsing and Translation:** The DBMS checks the syntax and semantics of the SQL query and converts it into an internal representation.
2. **Query Optimization:** The optimizer generates different possible execution plans and estimates their costs.
3. **Plan Selection:** It selects the plan with the lowest estimated cost based on factors such as CPU usage, disk I/O, indexes, and number of rows.
4. **Query Execution:** The selected execution plan is executed by the query execution engine to produce the result.

For example, if a query searches for a student using Student_ID, the optimizer may choose an index scan instead of scanning the entire table because it requires less I/O.

Q2. What is the difference between heuristic optimization and cost-based optimization? When would you use each?

Heuristic optimization uses predefined rules to improve a query. For example, applying selection and projection operations early can reduce the amount of data processed.

Cost-based optimization generates multiple execution plans and calculates their estimated costs using factors such as disk I/O, CPU time, indexes, and number of records. It selects the plan with the lowest estimated cost.

Difference:

- Heuristic optimization → rule-based and faster.
- Cost-based optimization → statistics-based and usually produces a better plan.

I would use heuristic optimization for simple queries where quick optimization is required. Cost-based optimization is preferred for complex queries involving multiple joins and large tables.

Q3. How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.

In a real project, I would first create the database and tables. Then I would configure the database connection details such as URL, username, and password. The application establishes a connection using a database connectivity API. After that, SQL queries are executed using statements or prepared statements, results are processed, and finally the connection is closed.

JDBC stands for Java Database Connectivity. It is specifically designed to connect Java applications with databases.

ODBC stands for Open Database Connectivity. It is a general database connectivity standard that can be used by applications written in different programming languages.

For example, a Java-based banking application can use JDBC to connect to a MySQL database and retrieve customer account information.

Q4. What are ACID properties? Give a real-world example where each property is critical.

ACID properties ensure reliable and consistent transaction processing.

1. **Atomicity:** A transaction is completed fully or not at all.
Example: In a bank transfer, money must be deducted from one account and added to another. If one operation fails, the complete transaction is rolled back.
2. **Consistency:** A transaction must take the database from one valid state to another valid state.
Example: A bank account should not violate its defined constraints after a transaction.
3. **Isolation:** Concurrent transactions should not interfere with each other.
Example: Two people should not be able to incorrectly withdraw the same available balance.
4. **Durability:** Once a transaction is committed, its changes remain permanently stored even after a system failure.
Example: After a successful online payment, the payment record should not disappear after a power failure.

Q5. Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?

Concurrency control manages multiple transactions executing at the same time.

- **Dirty Read:** A transaction reads data written by another transaction before it is committed. If the first transaction rolls back, the data read was incorrect.
- **Lost Update:** Two transactions update the same data, and one update overwrites the other.
- **Unrepeatable Read:** A transaction reads the same data twice and gets different values because another transaction modified it between the two reads.

Two-Phase Locking (2PL) solves these problems by controlling access using locks.

2PL has two phases:

1. **Growing phase:** The transaction can acquire locks but cannot release them.
2. **Shrinking phase:** The transaction can release locks but cannot acquire new locks.

By controlling when transactions can read or modify data, 2PL ensures conflict-serializable schedules and reduces concurrency problems.

Q6. What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.

A **deadlock** occurs when two or more transactions wait for each other indefinitely.

For example, Transaction T1 holds a lock on resource A and waits for B, while T2 holds B and waits for A. Neither transaction can continue.

Deadlock Detection:

The DBMS creates a **Wait-For Graph**. Transactions are represented as nodes and waiting relationships as directed edges. If the graph contains a cycle, a deadlock exists.

Prevention strategies:

1. **Wait-Die:** An older transaction may wait for a younger transaction, while a younger transaction may be rolled back.
2. **Wound-Wait:** An older transaction can force a younger transaction to roll back instead of waiting.

Another strategy is to use a timeout and abort a transaction if it waits for too long.

Q7. Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?

Immediate Update allows database changes to be written to disk before the transaction commits. Therefore, both UNDO and REDO operations may be required during recovery.

Deferred Update does not write the database changes to disk until the transaction commits. Therefore, recovery generally requires REDO of committed transactions.

Comparison:

Immediate Update

Deferred Update

Changes can be written before commit	Changes are written after commit
UNDO and REDO may be required	Mainly REDO is required
More flexible for database systems	Simpler recovery
Suitable when early writing is useful	Suitable when simpler recovery is preferred

For example, immediate update can be preferred in systems where efficient buffer management is important, while deferred update is useful when simple recovery logic is desired.

Q8. Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?

Shadow Paging is a recovery technique that maintains two page tables: a shadow page table and a current page table.

The shadow page table points to the original database pages. When a transaction modifies a page, a new copy of that page is created instead of modifying the original page. After successful completion, the current page table becomes the new stable version.

Advantages:

- No log records are required.
- Recovery can be fast because the system can use the shadow page table.
- It provides a simple recovery mechanism.

Limitations:

- It can cause fragmentation.
- Maintaining page tables can require additional overhead.
- It is less suitable for very large and frequently updated databases.

Compared with log-based recovery, shadow paging is simpler, while log-based recovery is generally more flexible and widely used in large database systems.

Q9. What is a serializability schedule? How do you test for conflict serializability using a precedence graph?

A **serializable schedule** is a concurrent schedule whose result is equivalent to the result of some serial execution of the same transactions.

To test **conflict serializability**, we use a **precedence graph**.

Steps:

1. Create one node for each transaction.

2. Examine conflicting operations such as Read-Write, Write-Read, and Write-Write on the same data item.
3. If transaction T_i performs the conflicting operation before T_j , create an edge from T_i to T_j .
4. Check the graph for cycles.

If the precedence graph is **acyclic**, the schedule is conflict serializable.

If the graph contains a **cycle**, the schedule is not conflict serializable.

For example, if the graph contains $T_1 \rightarrow T_2 \rightarrow T_3$ and there is no edge returning to T_1 , it is acyclic and therefore conflict serializable.

Q10. Describe Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?

Basic 2PL: A transaction has a growing phase in which it obtains locks and a shrinking phase in which it releases locks. It guarantees conflict serializability.

Strict 2PL: All exclusive/write locks are held until the transaction commits or aborts. This prevents other transactions from reading or modifying uncommitted data and simplifies recovery.

Rigorous 2PL: Both shared/read locks and exclusive/write locks are held until the transaction commits or aborts.

Comparison:

- Basic 2PL → ensures serializability but may allow cascading rollback.
- Strict 2PL → holds write locks until commit and simplifies recovery.
- Rigorous 2PL → holds all locks until commit or abort.

Strict 2PL is commonly used because it provides conflict serializability, prevents dirty writes, and makes recovery easier while still allowing reasonable concurrency.